

A Diagnostic Pivot: Paroxysmal Nocturnal Hemoglobinuria Mimicking Refractory Immune Thrombocytopenia in an Elderly Woman With Autoimmune Comorbidities

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INTRODUCTION

Immune thrombocytopenia (ITP) is an acquired autoimmune disorder characterized by isolated thrombocytopenia due to immune-mediated platelet destruction. While common in older adults, ITP is usually responsive to corticosteroids, IVIG, or thrombopoietin receptor agonists. **Lack of response should prompt reconsideration of the diagnosis.**

Paroxysmal nocturnal hemoglobinuria (PNH) is a clonal stem cell disorder caused by PIGA mutations and loss of glycosylphosphatidylinositol (GPI)-anchored proteins CD55 and CD59, which predisposes cells to complement-mediated lysis and manifests with hemolysis, cytopenias, and thrombosis.

CASE DESCRIPTION



Patient Demographics

- 82-year-old woman
- Medical History: Sjögren's, Hashimoto's, Osteoporosis



Presentation

- Fatigue and Recurrent Epistaxis
- Severe thrombocytopenia (Platelets $9 \times 10^9/L$)
- 1-year of gradually declining platelet counts blamed on Denosomab



Initial Treatment Response

- Did not respond to corticosteroids, IVIG, or Romiplostim over two months

LABORATORY FINDINGS

Test	Result	Clinical Significance
Hemoglobin	9.2 g/dL	↓ Mild anemia
Platelets	$9 \times 10^9/L$	↓↓ Severe thrombocytopenia
White Blood Cells	$4.2 \times 10^9/L$	No leukopenia
Reticulocyte Count	3.7%	↑ Appropriate marrow response to hemolysis
LDH	564 U/L	↑ Intravascular Hemolysis
Haptoglobin	<10 mg/dL	↓ Intravascular Hemolysis
Direct & Indirect Coombs (DAT)	Negative	Excludes autoimmune & alloimmune hemolysis
Peripheral Smear	No schistocytes, no spherocytes	Excludes microangiopathic and hereditary membrane causes
PT/INR & aPTT	Normal	Rules out DIC, MAHA, or consumptive coagulopathy

DIAGONSTIC STUDIES

BONE MARROW BIOPSY

Normocellular marrow with preserved hematopoiesis

FLOW CYTOMETRY

Evidence of loss of several GPI linked markers, including CD59, CD24, & CD14
FLAER-based assay: 31.12% of granulocytes, 28.31% of monocytes, and 12.12% of erythrocytes

TREATMENT OUTCOME

RAVULIZUMAB

$9 \times 10^9/L$
BASELINE
PLATELETS

$42 \times 10^9/L$
POSTTREATMENT
PLATELETS

3 WEEKS
TIME FOR
IMPROVEMENT

DISCUSSION

- This case shows how PNH may be mistaken for ITP when thrombocytopenia is the dominant feature. The lack of response to ITP-directed therapy and evidence of hemolysis were key to reconsidering the diagnosis.
- **Flow cytometry should be performed in patients with unexplained cytopenias when bone marrow morphology is preserved with persistent hemolysis.** PNH may present as isolated thrombocytopenia due to preferential complement-mediated destruction of platelets, complicating distinction from ITP.
- Autoimmune conditions may further obscure the diagnosis, given their association with secondary ITP.
- Early recognition of PNH is critical because complement inhibition alters disease course by reducing hemolysis, decreasing transfusion needs, and preventing thrombosis.
- In older patients, accurate diagnosis avoids unnecessary immunosuppression and allows timely initiation of targeted therapy.

KEY CLINICAL INSIGHTS

Hemolysis Recognition - Early identification of hemolysis features not consistent with ITP

Anemia with Minimal Bleeding - Hemoglobin 9.2 with only mucosal bleeding suggests hemolysis

Classic Hemolysis Markers - Low haptoglobin, elevated bilirubin, elevated LDH

Treatment Resistance - Lack of response to standard ITP therapy should prompt reconsideration

STRENGTHENING THE APPROACH

Clarify Role of Anemia and Hemolysis

- In isolated ITP, hemoglobin is typically normal unless bleeding is significant
- Hemoglobin of 9.2 with only mucosal bleeding, low haptoglobin, elevated bilirubin, and elevated LDH suggested hemolysis early on
- These features were not consistent with ITP

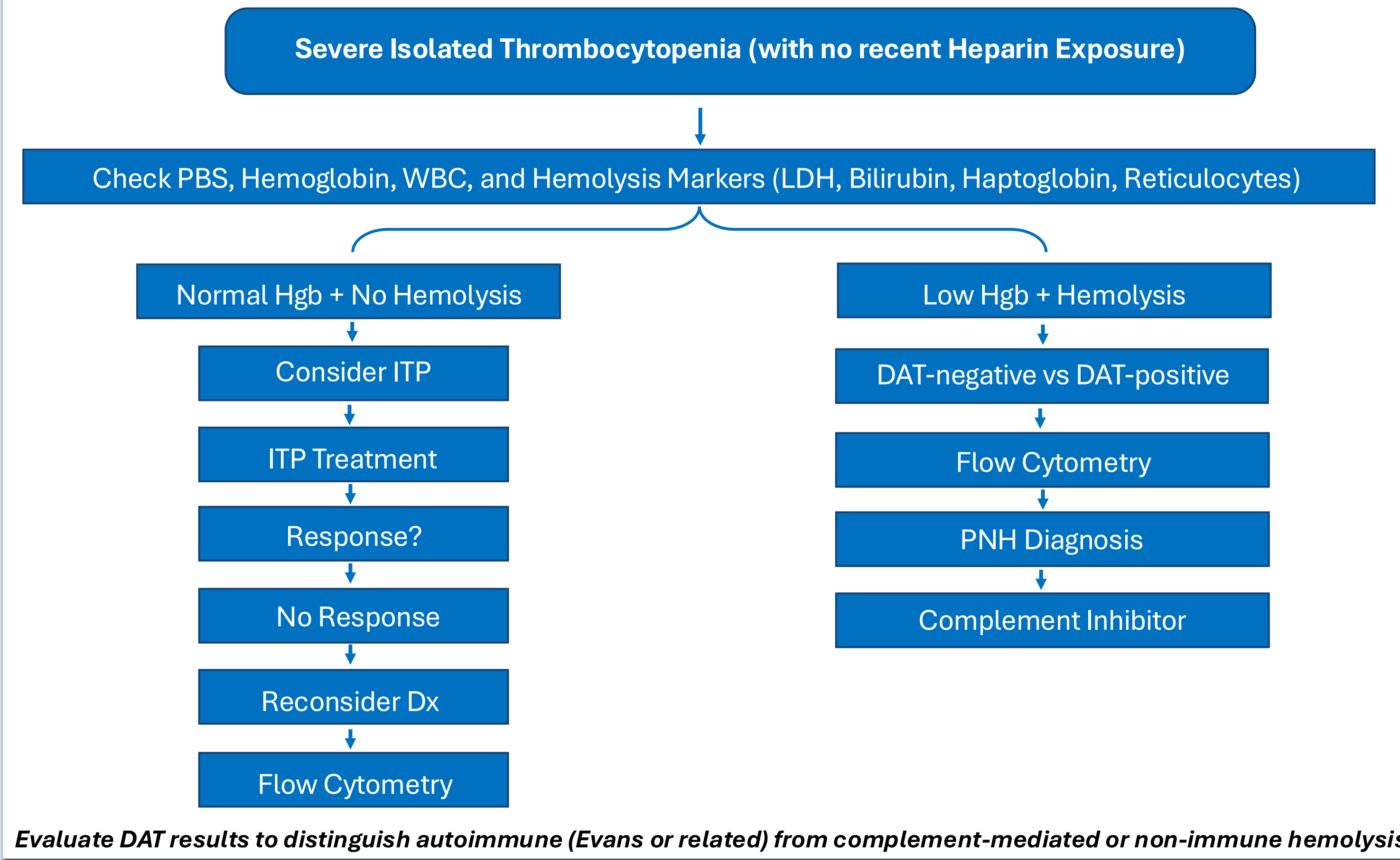
Refine Differential Diagnosis

- Consider Evans syndrome when both thrombocytopenia and anemia are present
- Unlikely in this case given negative DAT
- Demonstrates balanced diagnostic thought process

Address Timing of Work-up vs. Empiric Therapy

- Empiric ITP-directed therapy is reasonable with severe thrombocytopenia
- Persistent anemia and hemolysis in the absence of major bleeding should prompt earlier evaluation
- Consider flow cytometry for PNH earlier in the diagnostic process

DIAGNOSTIC ALGORITHM



DIFFERENTIAL DIAGNOSIS COMPARISON

Feature	ITP	PNH	Evans Syndrome
Thrombocytopenia	Yes	Yes	Yes
Anemia	Only with bleeding	Yes (hemolytic)	Yes (hemolytic)
Hemolysis markers	Normal	Elevated	Elevated
Direct Antiglobulin Test	Negative	Negative	Positive
Response to steroids/IVIG	Usually responsive	Non-responsive	Variable
Flow cytometry	Normal	GPI-deficient cells	Normal
Treatment	Steroids, IVIG, TPO-agonists	Complement inhibitors	Immunosuppress

WHEN TO SUSPECT PNH

Top Priority

Secondary Indicators

Supporting Features

- Unexplained cytopenias with preserved bone marrow morphology
- Persistent hemolysis markers
- Treatment-resistant thrombocytopenia
- Negative direct antiglobulin test
- Reticulocytosis
- Normal iron studies
- History of autoimmune conditions

CONCLUSION

This case highlights the importance of reconsidering the diagnosis when patients do not respond to standard ITP-directed therapy. The presence of hemolysis markers, even with thrombocytopenia as the dominant feature, should prompt evaluation for PNH. Early recognition allows for timely initiation of complement inhibitor therapy, which significantly improves outcomes by reducing hemolysis, decreasing transfusion needs, and preventing thrombotic complications.